# WHAT TWO SEASONS OF TRAINING AND PERFORMANCE DATA CAN TELL A PERFORMANCE DEPARTMENT

Load, wellness and injury monitoring in two elite women's football teams (SoccerMon, 2020-2021)

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## SUMMARY FOR COACHES AND STAFF

**In one sentence**: training load does impact how players feel, but each session's effect is small; wellness scores gave no prediction or warning of injury; and the biggest problem in the data is that players don't report consistently.

**Five Findings and Action to Take:**

1. **Load swings significantly throughout the season.** The quads monthly loads nearly tripled from the winter break in Dec 2020 to the pre-season build up in March 2021, and players were above the ACWR danger line (1.5) on 1 in 7 training days. Action: plan graded returns after long breaks and use the weekly monitor to catch sikes before they cause issues.

2. **Harder days make players feel slightly more fatigued the next day.** The impact is real and consistent, but small: a double length session moves reported next-day fatigue by approximately a tenth of a point on a 1-5 scale. Action: assess wellness scores to spot players drifting below their normals over a week, rather than reacting to a single session.

3. **Wellness scores gave no prior waring before injuries.** Players felt no worse in the week prior to injury than in the two weeks before that (days 8-21 prior). Injuries were about twice as common when ACWR was above 1.5, but with a sample size of just 29, this can't be proven. Action: treat flags as a prompt for conversation, not a prediction of injury.

4. **Knees are the problem area, and injury records understate time out.** Knees accounted for 38% of injury reports and 70 out of 76 major reports. One player reported a major knee injury and then recorded zero load for a week; whereas another reported an injury for 163 days whilst training normally. Action: record each players daily availability, not just symptoms.

5. **Data quality is limited by reporting compliance.** Players completed 67% of possible wellness forms (Team A – 75%. Team B – 56%). Team B recorded only 4 of 29 total injuries despite making up almost half of the cohort., which suggest injuries went unreported. Action: make the check in part of the daily routine and chase missing reports; the dashboard lists the lowest reporting players.

**How this was built:** two seasons of real data from two elite women's teams (SoccerMon), cleaned in Excel/Power Query, modelled in Power BI and tested in R. A live dashboard and automated weekly report are also included.

# 1. INTRODUCTION

Elite clubs collect training load and wellness data every day, but the data is only worth collecting if it can be utilised to inform decisions. Session rating of perceived exertions (sRPE), a players self-reported effort rating, multiplied by sessions duration, is the most widely used measure of internal load (Foster et al., 2001). The acute chronic workload ratio (ACWR), which compares the last 7 days load with the last 28 days, has been linked to risk of injury (Gabbett, 2016), although its statistical basis has been strongly criticised (Impellizeri et al., 2020). Daily questionnaires focusing on wellness are used alongside load, with the aim of highlighting building fatigue before it becomes a problem.

The project used two seasons of data from two elite women's football teams to answer 3 main questions:

**Q1 – Does a higher training load predict worse wellness the next day?**

**Q2 – Do wellness scores get worse in the weeks before an injury?**

**Q3 – Are injuries more common when ACWR is high?**

Alongside the analysis, the project produced a reusable monitoring system: a clean data model, a five-page Power BI dashboard, a four-page R dashboard and an automated weekly squad report.

# 2. DATA

The data came from SoccerMon (Midoglu et al., 20204), an open dataset of daily self-reported data from two elite women's football teams in Norway throughout the 2020 and 2021 seasons, collecting through the PmSys app. The players are anonymised by ID and the data was published under a CC BY 4.0 license. Ths dataset covers 50 players over 731 days: including 16,265 training sessions, 16,992 wellness reports, 162 injury reports and 15 illness reports. Wellness items were scored from 1-5 with higher scores representing better wellness.

This project does not use GPS data (published separately), and the dataset has no match performance data, nutrition data or record of player availability. Injury information is self-reported symptom, not a medical diagnosis.

**Table 1. Sample by team, 2020-2021 seasons.**

| Team | Players | Training Days | Mean load per training day (AU) | Wellness reports | Injury Episodes |
|---|---|---|---|---|---|
| A | 27 | 9,599 | 482 | 11,020 | 25 |
| B | 23 | 4,592 | 540 | 5,972 | 4 |
| Total | 50 | 14,191 | 501 | 16,992 | 29 |

# 3. METHODS

## 3.1 DATA PIPELINE

The raw data was organised into 15 wide CSV files, with one column per player and one row per day. These were reshaped in Power Query into a one long table with a single row per player per day (411,359 values into 36,550 player-days, 50 players x 731 days). Load for each session was calculated as sRPE (RPE x session duration in minute, in arbitrary units, AU) and summed per day. ACWR was calculated as the 7-day load divided by the weekly average of the 28-day load and checked against the ACWR values supplied in the original dataset for accuracy. Weekly monotony (mean daily load/ its standard deviation) and strain (weekly load x monotony) follower Foster (1988). The cleaned tables were loaded into Power BI as a star schema with one player table and one date table.

## 3.2 INJURY EPISODES

Injury reports are daily, so one injury can produce many reports and some reports list multiple body areas. The 162 injury reports were split into 306 body-area rows (76 flagged as major), then grouped into episodes: reports for the same player were treated as a single episode unless there was a gap of more than 14 days between them. A shorter gap split single injuries into several episodes, whereas a longer ap merged injuries that were clearly separate. This gave 29 unique injury episodes. Episode length is the total number of days a player reported the injury, which importantly is not the same as time lost from training (Section 4.5).

## 3.3 WELLNESS SCORES

Players approaches to reporting using the 1-5 scale can differ, so each wellness item was converted to a z-score relative to that players own mean and standard deviation. A score of -1 means one standard deviation below the players normal. A 7-day wellness index was the average z-score across each of the wellness metrics (fatigue, soreness, sleep quality and stress), over the previous 7 days. Averaging over a week reduces the spread of the index to about 0.42 standard deviations, so a threshold of -0.5, rather than the usual -1, was used to flag a player as below their normal.

## 3.4 DATA QUALITY CHECKS

Compliance was calculated as wellness reports divided by active player-days (days between each player's first and last recorded activity), giving 25,529 active player days. Checks covered duplicate rows , values outside the 1-5 scale, impossible durations, text in numeric columns and gaps in each player's record. All decisions are listed in Appendix A.

## 3.5 STATISTICAL ANALYSIS

All statistical analysis was completed in R 4.5.1. For Q1, linear mixed effects models (lme4, with ImerTest for p-values) predicted the next-day fatigue and soreness from that day's load (per 100 AU), adjusting for the same-day score, with a random intercept for each player so that each player was compared against their own baseline. The analysis used 11,980 player-days with both a next-day training report and training load. Robustness was checked by removing the top 1% of loads and by fitting the model for each of the teams separately. For Q2, fatigue z-scores in the final 7-days before each injury were compared with days 8-28 before it, using a mixed model with a random intercept for each episode, and a pried t-test on episode averages as a simpler check. For Q3, injury rates (per 1000 player-days) were calculated for four ACWR bands (below 0.8, 0.8-1.3, 1.3-1.5, above 1.5), with exact Poisson confidence intervals, and the above 1.5 band was compared with 0.8-1.3 as a rate ratio. Results are reported as estimates with 95% confidence intervals; $p < 0.05$ was treated as statistically significant.

# 4. RESULTS

## 4.1 LOAD ACROSS THE SEASON

Squad load varied widely across the two seasons (Figure 1). Monthly quad load raged from 187,890 AU in December 2020 to 548,610 AU in March 2021, a 2.9-fold difference between the winter break and the peak of pre-season. ACWR was above the 1.5 threshold on 14% of training days and below 0.8 on another 14%. High ACWR was more common in Team B (17.8% of training days) than Team A (12.1%). Mean load per training day was 482 AU in Team A and 540 AU in Team B.

***Figure 1. Squad overview dashboard: weekly load by team and monthly squad totals, 2020–2021.***

This means sharp rises in load were common, especially after breaks, which is where a weekly monitor is most useful.

## 4.2 LOAD AND NEXT-DAY WELLNESS

Higher training load predicted slightly worse wellness the next day (Figure 2). Each additional 100 AU of load predicted next-day fatigue 0.034 points worse (95% Cl 0.031 to 0.038, p<0.001) and next day soreness 0.039 points worse (95% Cl 0.035 to 0043, p<0.001). The raw data illustrated the same pattern: average next-day fatigue fell from 3.13 after 1-200 AU sessions to 2.83 after sessions over 800 AU (Figure 3). The result held when the top 1% of loads were removed (0.037 per 100 AU) and in each team separately (Team A 0.030, Team B 0.045).

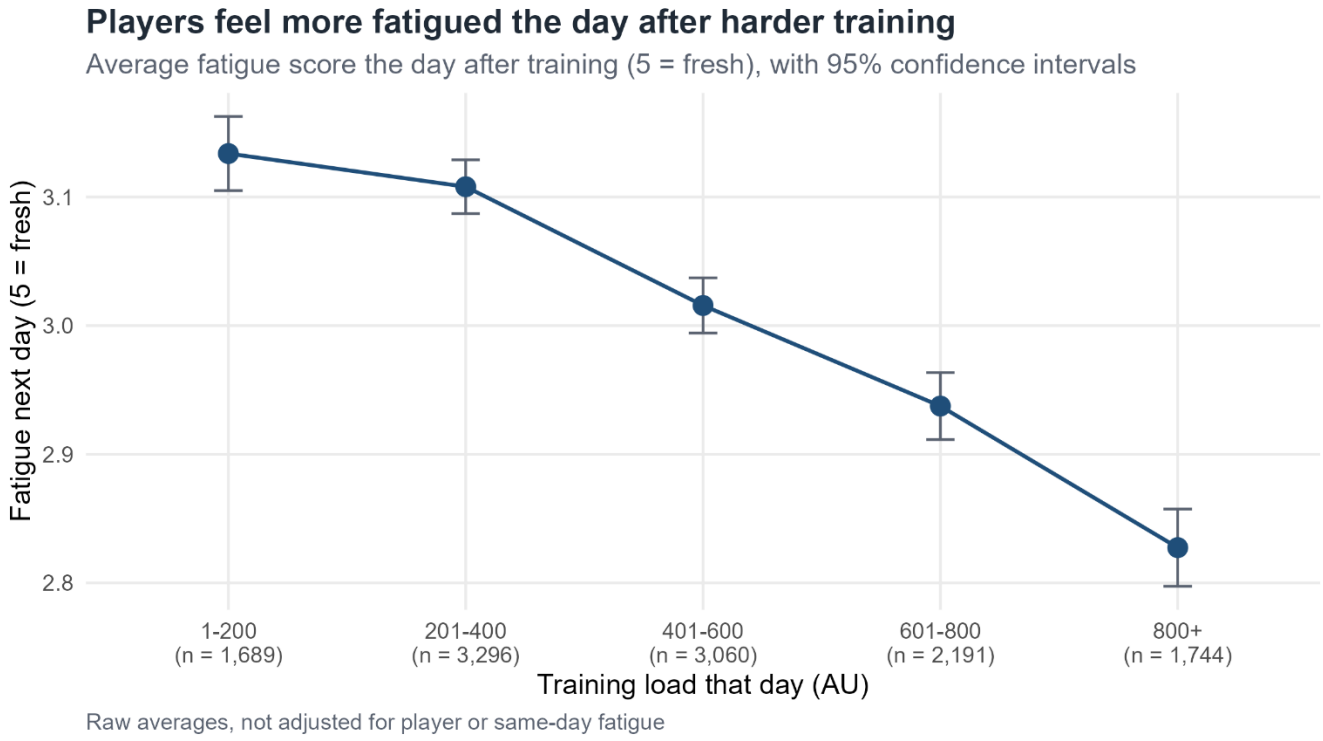



***Figure 2. Predicted next-day fatigue and soreness by training load, from mixed-effects models (shaded band = 95% confidence interval).***

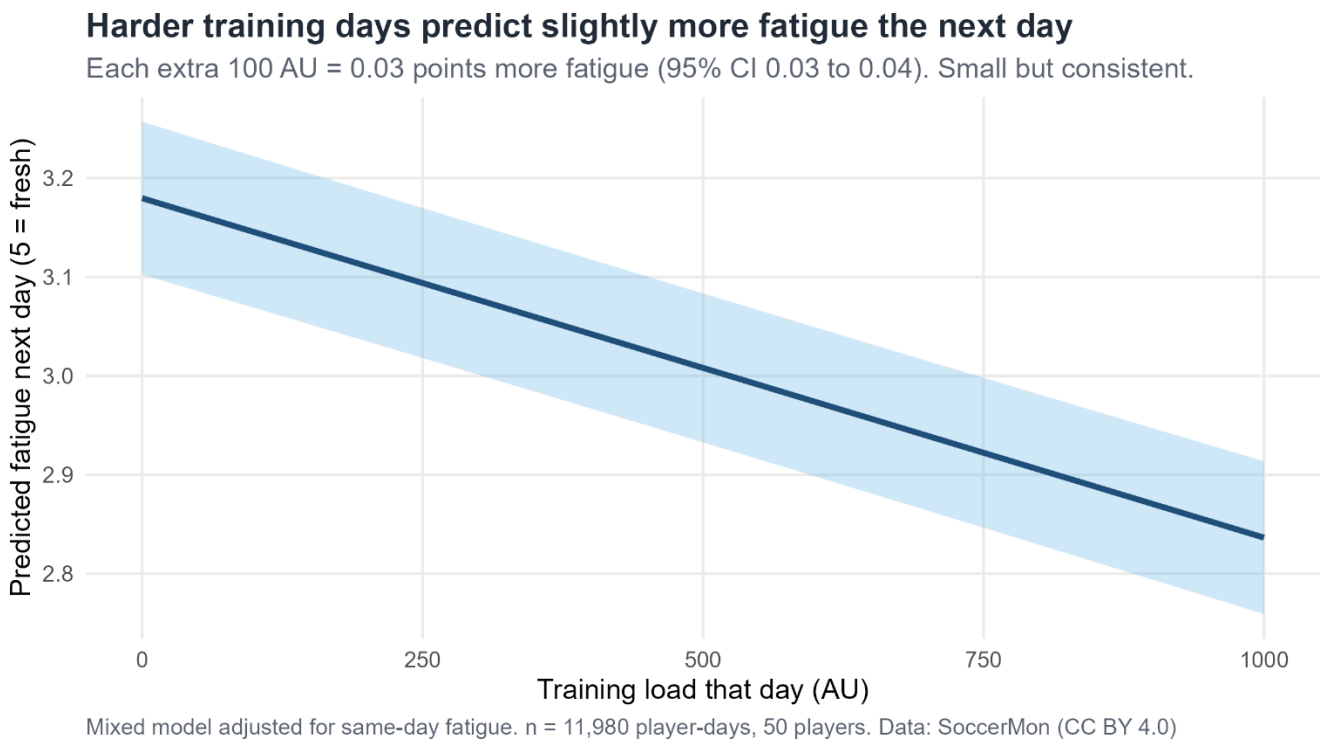



***Figure 3. Average next-day fatigue score by training load band (raw data, before modelling).***

The effect is reliable but small. A 600 AU session compared with a 300 AU sessions predicts next-day fatigue only about 0.10 points worse, and soreness 0.12 points worse, on a 1-5 scale. That is far smaller than the normal day-to-day variation in a player's scores.

## 4.3 WELLNESS BEFORE INJURY

Wellness did not get worse in the week before injury (Figure 4). In the mixed model, fatigue z-scores in the final 7 days before injury were 0.23 standard deviations higher (better) than in the 8-28 days before it (95% CI -0.03 to 0.49, p=0.09). The paired t-test on the 17 episodes with data in both windows gave the same estimate (+0.23, 95% CI -0.18 to 0.64, p=0.26).

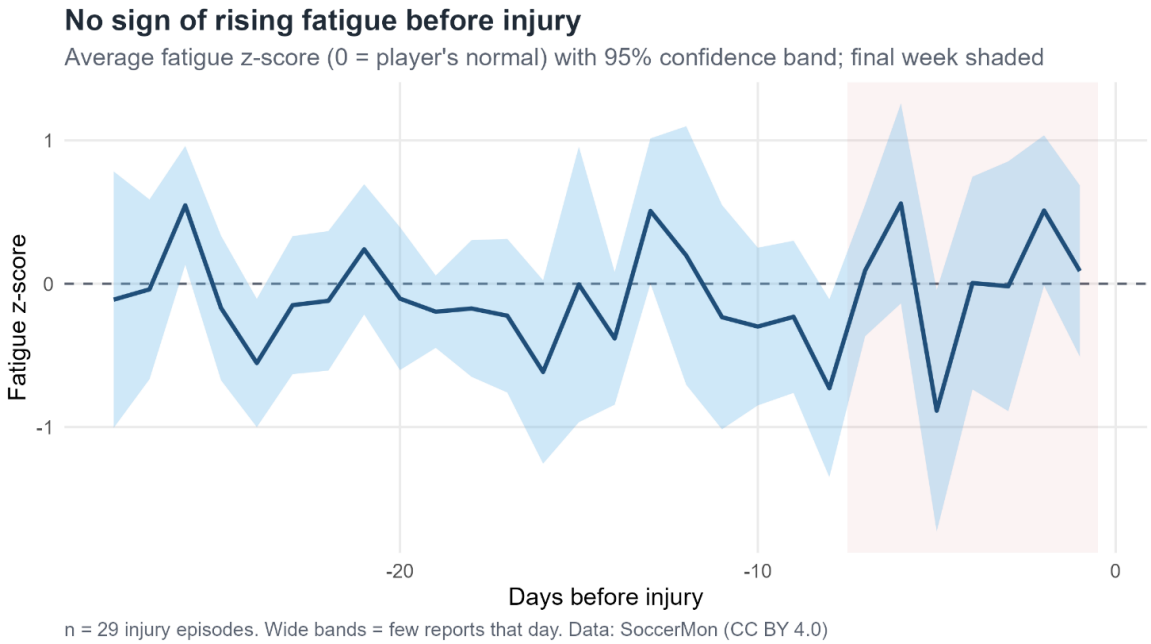


***Figure 4. Fatigue z-score in the 28 days before injury onset (0 = the player's normal).***

If anything, players felt slightly fresher just before getting injured, the opposite of an early warning, although the difference was not statistically significant.

## 4.4 ACWR AND INJURY

Injury rates were highest when ACWR was above 1.5 (Table 2, Figure 5). Of the 29 episodes, only 20 had a valid ACWR on the day of onset; the rest started before a player had recorded 28 days of load data. The injury rate above 1.5 was 1.95 per 1000 player-days, compared with 0.83 in the 0.8-1.3 band, a rate ratio of 2.5 (95% CI 0.70 to 7.15, p=0.11).

***Table 2. Injury rate by ACWR band (20 of 29 episodes with a valid ACWR).***

| ACWR band | Player-days | Injuries | Rate per 1000 | 95% CI |
|---|---|---|---|---|
| Below 0.8 | 4301 | 3 | 0.70 | 0.14 to 2.04 |
| 0.9-1.3 | 12091 | 10 | 0.8 | 0.40 to 1.52 |
| 1.3-1.5 | 1630 | 1 | 0.61 | 0.02 to 3.42 |
| Above 1.5 | 3081 | 6 | 1.95 | 0.71 to 4.24 |

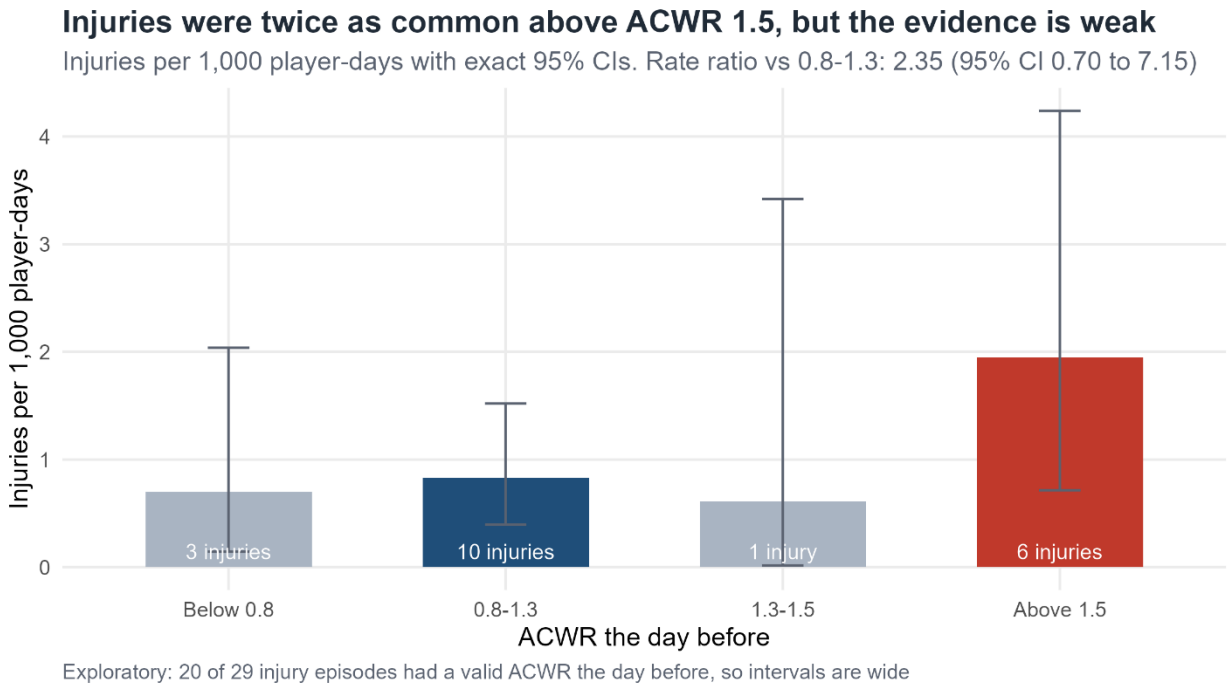



***Figure 5. Injury rate per 1,000 player-days by ACWR band (error bars = 95% confidence intervals).***

The direction matches previous research, but with 6 injuries in the high band the confidence interval is very wide and includes no difference at all. This dataset cannot confirm or rule out a link.

## 4.5 INJURY PROFILE

The 162 injury reports became 306 unique body-area rows and 29 episodes (Team A – 25, Team B 4), 6 of which were classified as major (Figure 6). Knees made up 116 of 306 body-area reports (38%) and 70 of 76 major reports. Players reported injury on a total of 404 days.

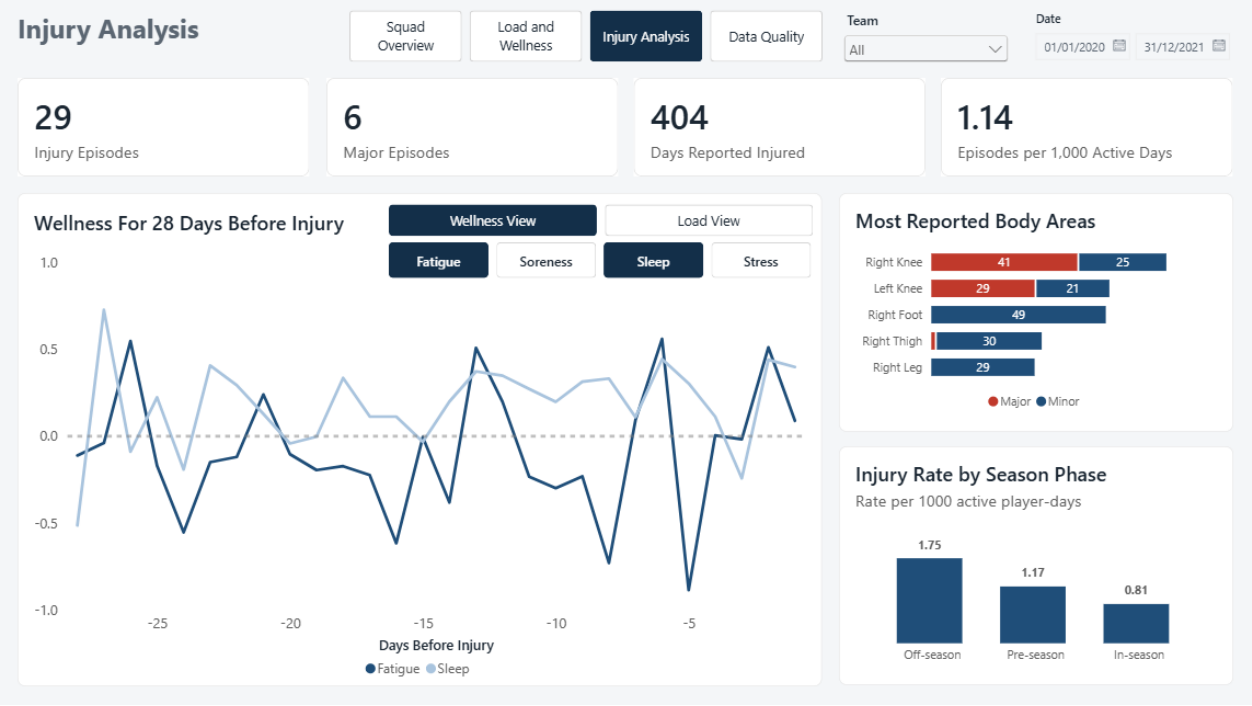



***Figure 6. Injury dashboard: episodes, body areas and wellness around injury.***

However, days reported is not the same as tie lost due to injury. Player A-74af reported a major right-knee injury once, on 25/03/2021 then trained at zero load for a week and took four weeks to rebuild to her previous load (2480 AU per week before; then 0, 1260, 2530, 2810). Her record shows a one-day episode. Player A-4051 had a 163-day major episode while training around 3100 AU per week. Without a record of availability, the data cannot give a value to the training time lost.

## 4.6 DATA QUALITY

66.6% of possible wellness reports were completed: 74.7% in Team A and 55.5% in Team B (Figure 7). Individual compliance ranged from 12% to 99%, and 14 players reported less than half the time, 10 of which were from Team B. Team B made up 46% of players and 42% of training days but recorded just 4 of the 29 injury episodes (0.37 per 1000 training days, against 1.69 in Team A). Only 15 illness reports came from the whole squad, all from 4 players.

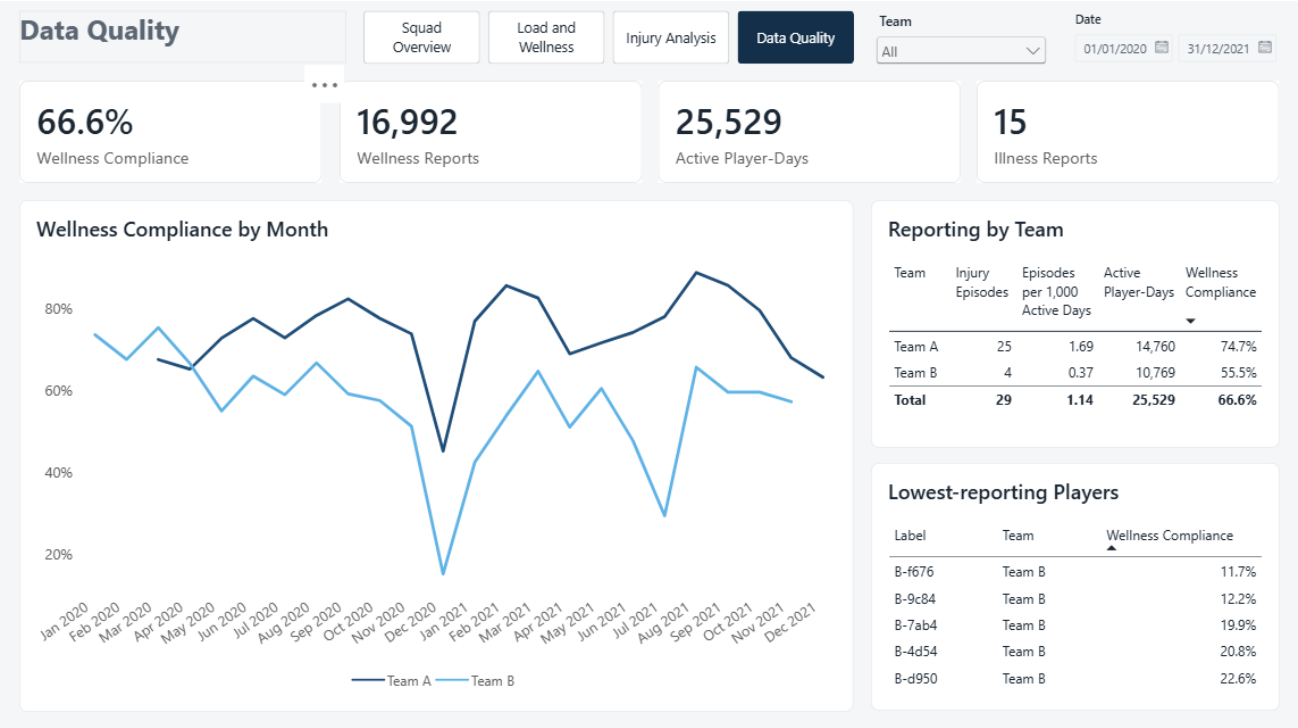



***Figure 7. Data quality dashboard: wellness compliance by team and by player.***

These gaps mean the missing data is not random: the players with poor reporting are likely to be the ones that staff know least about.

# 5. DISCUSSION

Together, the results suggest that load and wellness data are more useful for monitoring trends than for predicting future injury. Load affected next-day wellness in a consistent direction, but by about a tenth of a point per 300 AU, so a single low score after a hard session is expected rather than a warning sign. Wellness did not fall before injury, which suggests many of these injuries were acute (such as injuries in contact or landing) rather than a result of accumulating fatigue. Injury rates were higher above ACWR 1.5, in line with Gabbett (2016), but the evidence is weak and the ACWR itself has known issues (Impellizzeri et al., 2020), which is why the dashboard lets staff set their own limits rather than fixing them.

The clearest finding is about the data rather than the players. Team B's low reporting of both wellness and injuries makes every other result less certain, because it is likely that the least-monitored players are also those whose injuries went unrecorded.

## 5.1 LIMITATIONS

- All data is self-reported, and compliance varied from 12% to 99% between players.
- 29 injury episodes, and only 20 with valid ACWR, give very little statistical power for Q2 and Q3.
- Episode length reflects reporting, not time lost from training.
- Injury reports are symptoms, not medical diagnoses, and Team B probably under-reported.
- The thresholds (ACWR 0.8 and 1.5, wellness -0.5) are convention, not proven cut-offs.
- The data is observational: it shows correlations, not causation.
- Two seasons from one club setting; results may not apply to academy or men's football.
- External load (GPS) was not included.

# 6. RECOMMENDATIONS

| Recommendation | Who | How to Monitor Impact? |
|---|---|---|
| Plan graded returns after breaks; review ACWR spikes in weekly report | S&C | Fewer training days above ACWR 1.5 |
| Judge wellness over a week against each player's normal, not after single sessions | S&C, Physio | Flags based on the 7-day index |
| Use flags to start a conversation, not to predict injury | All Staff | Flags followed up within 24 hours |
| Record daily availability (full, modified, unavailable) and time lost | Physio | Time lost recorded for every episode |
| Make the check-in part of the daily routine and chase missing reports the same day | Head of Performance | Compliance above 80% in both teams |

### 6.1 NEXT STEPS

- Add GPS data to link external load with sRPE
- Collect data across more seasons so Q2 and Q3 have enough injuries to test properly
- Only build a predictive model once there is enough injuries and availability data to test it.

## 7. TOOLS AND OUTPUTS

Data was cleaned and organised in Excel and Power Query, modelled and visualised in Power BI (star schema, 46 DAX measures), and analysed in R (lme4, lmerTest, ggplot2). The R dashboard and weekly report were built with Quarto. All code and outputs are on GitHub.

## 8. REFERENCES

Foster, C. (1998) 'Monitoring training in athletes with reference to overtraining syndrome', *Medicine & Science in Sports & Exercise*, 30(7), pp. 1164-1168.

Foster, C. et al. (2001) 'A new approach to monitoring exercise training', *Journal of Strength and Conditioning Research*, 15(1), pp. 109-115.

Gabbett, T.J. (2016) 'The training-injury prevention paradox: should athletes be training smarter and harder?', *British Journal of Sports Medicine*, 50(5), pp. 273-280.

Impellizzeri, F.M. et al. (2020) 'Acute:chronic workload ratio: conceptual issues and fundamental pitfalls', *International Journal of Sports Physiology and Performance*, 15(6), pp. 907-913.

Midoglu, C. et al. (2024) 'A large-scale multivariate soccer athlete health, performance and position monitoring dataset', *Scientific Data*, 11, 553.

# APPENDIX A: DATA DECISIONS

Every decision made while cleaning, building and analysing the data.

***Table A1. Data and analysis decisions***

| # | Decision | Reason |
|---|---|---|
| **Combining and cleaning the files** | | |
| 1 | Renamed first column of every daily file to 'Date' before combining them | Source files name it inconsistently, which stops them combining |
| 2 | Read dates as day.month.year (German locale) | Source writes dates as dd.mm.yyyy with dots, which UK settings misread |
| 3 | Treated blank value as missing, not zero | Blank means a player didn't report that day, not that she scored 0 |
| 4 | Converted text in numeric columns to numbers, and set values that couldn't be converted to missing | Some columns loaded as text, which broke averages in Power BI and R |
| 5 | Corrected the typo 'soar throat' to 'sore throat' in the illness data | Spelling error in source data |
| **Players and active days** | | |
| 6 | Gave each player a short label: team letter + first 4 ID characters (e.g. A-d729) | Readable on charts while staying anonymous and unique |
| 7 | Counted a player as active on a day from her first to her last training day of training or wellness reporting (25529 active player days) | Compliance should only count days the player was actually part of the squad |
| 8 | Kept early 2020 data, but treated Team A vs Team B comparisons before mid-March 2020 with caution | Team A only started logging load in March 2020, so early differences reflect logging, not training |
| **Wellness** | | |
| 9 | Converted each wellness item to a z-score against that player's own average and spread | Players use the 1–5 scales differently, so comparing each player with herself is fairer than raw scores |
| 10 | Treated higher scores as better on every wellness scale, so a negative z-score means worse than normal | Confirmed from the data: all wellness scores rise together with readiness and sleep duration |
| 11 | Built a 7-day wellness index: the average of the fatigue, soreness, sleep quality and stress z-scores over the week | One bad night shouldn't trigger a flag; combining items gives a steadier picture |
| 12 | Flagged low wellness at a 7-day index below −0.5, replacing the earlier single-day threshold (fatigue z below −1.5) | Averaging over 7 days shrinks the spread to about 0.42 SD, so stricter thresholds almost never triggered |
| 13 | Did not fill in missing wellness (33.4% of active days); reported compliance as its own measure | Guessing values would invent data; how often players report is useful information for staff |
| 14 | Kept the 53 sleep entries under 4 or over 11 hours, without interpreting single nights | Short or long nights happen (travel, illness); z-scores and averages limit their effect |
| **Training load and ACWR** | | |
| 15 | Checked that session load equals RPE × duration in all 16,265 sessions; no changes needed | Confirms the load data is internally consistent before anything is built on it |
| 16 | Kept and flagged the 4 sessions with RPE 0 and the 2 with zero duration | They add zero load, so totals don't change; removing them would break the match with the published figures |
| 17 | Kept the 73 sessions over 150 minutes; noted as a limitation | Match days with warm-ups, extra time and recovery can genuinely run long; no evidence they're errors |
| 18 | Kept and flagged the 151 sessions in 75 groups sharing the same player, date, RPE and duration | Two identical sessions in a day can be genuine, and SoccerMon's published daily loads include them |
| 19 | Kept the 274 sessions on the 87 player-days with 3 or more sessions (up to 6 in a day) | Likely gym, pitch and recovery logged separately; realistic for elite players |
| 20 | Rebuilt ACWR the way SoccerMon calculates it: "coupled", 7-day average ÷ 28-day average, with the 7 days inside the 28 | To reproduce the published figures exactly; the method is debated by researchers, so it is noted as a limitation |

| | | |
|---|---|---|
| 21 | Accepted differences of up to about 0.005 between my ACWR and SoccerMon's | SoccerMon rounds its intermediate values to 2 decimals; not an error |
| 22 | Flagged ACWR above 1.5 as high and below 0.8 as low, and weekly monotony above 2 as high | Conventional thresholds from the literature (Gabbett, 2016; Foster, 1998) |
| 23 | Let staff change the ACWR limits with a range slider in Power BI, rather than fixing them | The thresholds are conventions, not proven cut-offs, and staff may want their own |
| **Power BI model and dashboard** | | |
| 24 | Built a star schema: the fact tables (daily data, sessions, injuries) linked to one Date table and one Players table | Date, team and player filters then work the same way on every page |
| 25 | Rounded ACWR to 2 decimal places before colouring it against the limits | Tiny floating-point differences put values such as 1.47 in the wrong colour band |
| 26 | Blanked a player's weekly load outside her first and last active weeks on the profile charts | Zeros before she joined or after she left looked like real drops in training |
| **Injuries and illness** | | |
| 27 | Split reports listing several body areas into one row per area (162 reports → 306 rows) | Each body area needs counting and filtering separately |
| 28 | Started a new injury episode at a player's first report, or after a gap of more than 14 days (306 rows → 29 episodes) | Players log an injury every day it continues; 14 days stops a short gap splitting one injury in two, while a new injury months later counts separately |
| 29 | Counted an episode as Major if any report in it was major | The most serious report best describes the injury overall |
| 30 | Labelled episode length "days reported", not "days injured" | It counts reports, not time lost: A-74af reported a major knee injury once, then didn't train for a week |
| 31 | Used illness for description only, with no statistical tests | 15 reports from only 4 players is too few to test |
| **Statistical analysis** | | |
| 32 | Q1: used training days (load above 0) followed by a wellness report the next calendar day (11,980 player-days) | Tests the effect of an actual session on how the player felt the following day |
| 33 | Q1: mixed models with a random intercept for each player, adjusting for the same-day score | Each player has many days of data and her own baseline; adjusting stops a player who already felt bad from driving the result |
| 34 | Q1: robustness checks without the top 1% of loads and for each team separately | Checks the result isn't driven by extreme sessions or by one team |
| 35 | Q2: compared the final 7 days before injury with days 8–28 before it | Compares the final week with that player's own recent build-up; 28 days matches the chronic load window |
| 36 | Q2: ran a mixed model on all daily reports (random intercept per episode) and a paired t-test on episode averages (17 episodes with data in both windows) | The model uses all the data; the paired test counts each injury once, as a simpler check |
| 37 | Q3: used the previous day's ACWR for each day | On the day of injury, that day's session (possibly the one where the injury happened) is already inside the ACWR |
| 38 | Q3: excluded days with no ACWR or an ACWR of 0, leaving 20 of the 29 episodes | The ratio can't be calculated without load in the previous 28 days, and an ACWR of 0 means the player had no recent training to compare |
| 39 | Q3: four ACWR bands (below 0.8, 0.8–1.3, 1.3–1.5, 1.5 and above), compared with 0.8–1.3 using exact Poisson rate ratios | 0.8–1.3 is the conventional "sweet spot"; exact methods suit small injury counts |
| **Scope** | | |
| 40 | Did not use the GPS data | Kept the focus on internal load and wellness, which every club collects; GPS is listed as a next step |

# APPENDIX B: GLOSSARY

***Table B1:Glossary of terms***

| Term | Meaning |
|---|---|
| ACWR (acute:chronic workload ratio) | The player's average daily load over the last 7 days divided by her average over the last 28 days (the 7 days sit inside the 28). Above 1 means she is doing more than she is used to. |
| Active player-day | A day between a player's first and last day of training or wellness reporting. Used as the denominator for compliance. |
| AU (arbitrary units) | The unit of sRPE load. It has no physical meaning on its own; it is only useful for comparing sessions and players. |
| Compliance | Wellness reports ÷ active player-days. Shows how consistently a player fills in the daily form. |
| Confidence interval (95% CI) | The range the true value is likely to fall in. If it crosses 0 (or 1 for a rate ratio), the data cannot rule out no effect. |
| Daily load | The sum of a player's sRPE across all her sessions that day. |
| Injury episode | One injury, built from a player's daily injury reports. A new episode starts after a gap of more than 14 days without a report. |
| Major injury | An episode in which the player marked at least one report as major in the reporting app. |
| Mixed-effects model | A regression model that gives each player (or episode) her own baseline, so repeated measurements from the same player are handled properly. |
| Monotony | Mean daily load ÷ the standard deviation of daily load over a week. High monotony (above 2) means little variation between hard and easy days. |
| p-value | How likely a result at least this large would be if there were really no effect. Below 0.05 is called statistically significant. |
| Player-day | One player on one calendar day; the basic row of the data model. |
| Rate ratio | One injury rate divided by another. 2.0 means twice as many injuries per player-day. |
| RPE (rating of perceived exertion) | The player's rating of how hard a session felt, from 0 (rest) to 10 (maximal). |
| sRPE (session RPE) | RPE × session duration in minutes. The measure of internal training load used throughout. |
| Strain | Weekly load × monotony. High strain means a heavy week with little variation. |
| Wellness z-score | A wellness score expressed relative to that player's own average and spread. 0 is her normal; −1 is one standard deviation worse than normal. |
| 7-day wellness index | The average of the fatigue, soreness, sleep quality and stress z-scores over the last 7 days. Below −0.5 is flagged. |

# APPENDIX C: FULL STATISTICAL RESULTS

Full output from the models and tests summarised in Section 4. For the wellness scales, a negative estimate means worse wellness (higher scores are better). Marginal $R^2$ is the variation explained by the fixed effects alone; conditional $R^2$ adds the differences between players (or episodes). Observations are the rows with complete data for that model.

***Table C1. Fixed effects from the mixed-effects models.***

| Model | Term | Estimate | 95% CI | p |
|---|---|---|---|---|
| Q1: next-day fatigue | Load (per 100 AU) | −0.034 | −0.038 to −0.031 | <0.001 |
| Q1: next-day fatigue | Same-day fatigue | 0.213 | 0.194 to 0.231 | <0.001 |
| Q1: next-day soreness | Load (per 100 AU) | −0.039 | −0.043 to −0.035 | <0.001 |
| Q1: next-day soreness | Same-day soreness | 0.359 | 0.342 to 0.376 | <0.001 |
| Q2: fatigue z before injury | Final 7 days (vs days 8–28) | 0.229 | −0.034 to 0.493 | 0.088 |

***Table C2. Model size and fit.***

| Model | Observations | Players / episodes | Marginal $R^2$ | Conditional $R^2$ |
|---|---|---|---|---|
| Q1: next-day fatigue | 10,966 | 50 players | 0.072 | 0.278 |
| Q1: next-day soreness | 10,969 | 50 players | 0.174 | 0.341 |
| Q2: fatigue z before injury | 373 | 20 episodes | 0.007 | 0.097 |

***Table C3. Q1 robustness checks (next-day fatigue).***

| Q1 fatigue model | Load effect (per 100 AU) | Observations |
|---|---|---|
| Main model | −0.034 | 10,966 |
| Top 1% of loads removed | −0.037 | 10,887 |
| Team A only | −0.030 | 7,536 |
| Team B only | −0.045 | 3,430 |

***Table C4. Other tests.***

| Test | Result |
|---|---|
| Q2: paired t-test (17 episodes) | Mean difference 0.226 (95% CI −0.183 to 0.636), t(16) = 1.17, p = 0.259 |
| Q3: exact Poisson rate ratio, ACWR 1.5 and above vs 0.8–1.3 | 2.35 (95% CI 0.70 to 7.15), p = 0.113; 6 injuries in 3,081 player-days vs 10 in 12,091 |

# APPENDIX D: SOFTWARE AND REPRODUCIBILITY

***Table D1: Software used.***

| Tool | Version | Used for |
|---|---|---|
| Microsoft Excel with Power Query | Microsoft 365 | Combining, reshaping and cleaning the raw files |
| Power BI Desktop | 2.159.1231.0 | Data model, DAX measures and the five-page dashboard |
| R | 4.5.1 | Statistical analysis and charts |
| RStudio | 2026.10.1 | Running the R code |
| Quarto | 1.10.18 | The R dashboard and the weekly report |
| R packages | readxl 1.6.0, dplyr 1.2.1, tidyr 1.3.2, ggplot2 4.0.3, lme4 2.1-0, lmerTest 3.2-1, broom.mixed 0.2.9.7, performance 0.19.0, ggeffects 2.3.2 | Analysis (analysis.R) |
| R packages (reports) | gt, ggrepel | Weekly report tables and labels |

### How to reproduce this analysis:

1. Download the SoccerMon subjective data from Zenodo (https://zenodo.org/records/10033832) and unzip it.
2. Open SoccerMon_Clean.xlsx, update the source folder in Power Query (Data → Get Data → Data Source Settings), then Data → Refresh All.
3. Open SoccerMon_Dashboard.pbix and click Refresh.
4. Open the RStudio project in the R folder and run analysis.R from top to bottom. It writes every table, chart and test result to the outputs folder.
5. Render the R dashboard with quarto render dashboard.qmd, and a weekly report with quarto render weekly_report.qmd -P week_end:2021-09-15 -P team:All.

All code, the cleaned workbook, the dashboard file and the outputs are in the GitHub repository: github.com/will-page-495/soccermon-load-monitoring